

Blockchain in the Quantum Era: Surveying Security Challenges and Post-Quantum Cryptography

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Abstract—Recent advances in quantum technology are impacting cryptographic primitives, affecting their security. In this paper, we will survey the impact of these technologies on blockchain and distributed ledgers, and analyze the post-quantum primitives available to restore their security guarantees, including the latest NIST PQC standards ML-DSA-44 and SLH-DSA-SHA2-128s. Our analysis reveals critical trade-offs in signature size, key storage, and consensus resilience, highlighting challenges for IoT and PoW blockchains. Results underscore the urgency of improvements to the standardized NIST PQC algorithms to mitigate quantum risks to the blockchain without compromising throughput and decentralization.

Index Terms—Quantum-secure blockchain, Post-quantum Ethereum, Cryptographic migration, NIST PQC standardization, PQC in EVM, Ethereum quantum resilience

I. INTRODUCTION

Distributed Ledger Technology (DLT) and Blockchain have progressed significantly introducing accountability, redundancy, and transparency in multiple applications. Forbes indicates that the market capitalization of cryptocurrencies has now gone past the two-trillion-dollar mark. Blockchain technology has been identified as one of the trend-setting technologies that are disrupting industries and questioning existing patterns in the contemporary world in recent years.

However, the advancements made in recent years in Quantum computing, have raised alarms over blockchain security [1]. Due to the algorithms proposed in [2] and [3], public key cryptography, which is one of the pillars of blockchain, together with hash functions are threatened. Despite the current limited computation abilities of quantum computers, they are becoming a real threat with further development towards enhanced computation abilities that will apply much pressure on the basic cryptographic algorithms used in blockchain systems. Table 1 summarizes the key vulnerabilities due to quantum computing and the possible PQC solutions. These reliable cryptographic systems are known as Post-Quantum, Quantum-Resistant, Quantum-Proof, or Quantum-Safe cryptosystems [4].

In connection with this challenge researchers and developers have been putting their efforts into constructing a Post Quantum Blockchain (PQB) [5], which should ensure the security and sustainability of blockchain networks in the realm of

quantum computing. It is crucial to first give a background to quantum computing before providing details on the PQB [1].

PQB aims to include Post-Quantum Cryptography (PQC) algorithms into the structure of the blockchain in order to render the network resistant to quantum computing threats. A large number of proposed PQC algorithms can be incorporated into blockchain systems, mainly categorized as multivariate, lattice, and code-based cryptography. The integration of these algorithms will make it possible to achieve a PQB network that can provide long-term security and guard the data against quantum threats.

The implementation of PQB poses several challenges that have to be overcome in the process. First of all, it is crucial to understand that the shift from current cryptographic algorithms to PQC requires careful planning and considerable coordination. This implies an extensive overhaul of an already complex blockchain space. Furthermore, the need to have standardized PQC algorithms becomes imperative to enable different blockchain networks to interoperate or communicate effectively.

Surprisingly, there is a large number of academic initiatives dedicated to applying PQC security to blockchain. Currently, no PQB algorithm offers a complete package of low computational complexity, short signature/hash sizes, small key sizes, fast execution, and low energy consumption. Achieving this balance is especially crucial for resource-constrained devices like those in the “Internet of Things (IoT)”. The development and incorporation of PQB technology are still in their nascent phases [6]. The collective efforts of the research community, blockchain developers, and industry stakeholders are actively directed towards identifying the most efficient and secure PQC algorithms tailored for blockchain systems. Notably, standardization entities like the “National Institute of Standards and Technology” (NIST) have organized competitions aimed at selecting the most promising PQC algorithms. As the progression of quantum computing continues at an accelerated pace, it is crucial for the protection and continuation of blockchain use over time to adopt PQB technology.

In this paper we will analyse the most important vulnerabilities caused by the development of quantum technologies. Then

TABLE I
OVERVIEW OF QUANTUM VULNERABILITIES AND PQC SOLUTION APPROACHES

Quantum Vulnerability & Challenge	PQC Solution & Approach
Vulnerability of public-key cryptosystems (e.g., ECDSA, RSA) to Shor's algorithm	Adopt lattice-based schemes (e.g., CRYSTALS-Dilithium and CRYSTALS-Kyber) that rely on hard lattice problems, thereby offering quantum resistance.
Hash function collision security reduction due to Grover's algorithm	Use stateless hash-based signature schemes (e.g., SPHINCS ⁺) with extended output lengths to counteract Grover's quadratic speedup.
Key management and scalability issues in large decentralized networks	Develop lightweight PQC solutions with compact keys/signatures to reduce on-chain storage and bandwidth overhead.
Consensus and transaction verification challenges	Employ PQC algorithms optimized for fast verification and energy efficiency that can be integrated into existing consensus mechanisms.

we survey current post-quantum cryptographic primitives. Finally, we discuss the current PQC standards and their impact on distributed ledger technologies.

II. CRYPTOGRAPHIC VULNERABILITY TO QUANTUM

Shor's algorithm [2] can resourcefully resolve discrete logarithm problems (DLP) through the factorization of large integers, jeopardizing cryptosystems based on these principles. Shor's algorithm has an exponential speed-up, rendering integer factorization and DLP-based systems entirely vulnerable. Grover search algorithm [3] allows for quadratically faster searches aimed at hash codes and solutions to NP-complete problems compared to classical algorithms. Grover's impact on hash functions is quadratic, making hash-based systems PQC by doubling the key size [6]. Grover's algorithm offers two distinct avenues for its application in targeting a blockchain. The first method revolves around the identification of hash collisions, which is followed by the subsequent replacement of entire blocks within the blockchain. To illustrate, this algorithm to detect hash collisions in hash functions with a security level of n bits necessitates the production of an output with 3 times the number of bits ($3 * n$ bits). This implies that numerous existing hash functions may prove inadequate for the PQC era. As a result, hash functions might need to augment their output sizes to uphold security standards. The second approach centres on harnessing Grover's algorithm to expedite the mining process within blockchains like Bitcoin. This acceleration primarily pertains to the swift generation of nonces, potentially facilitating the rapid regeneration of complete blockchains, thereby posing integrity concerns.

ECDSA relies on the DLP over ECDLP for security, but both ECDLP and integer factorization (used by RSA) can be solved efficiently by quantum computers. The widely used elliptic-curve digital signatures algorithm (ECDSA) plays a crucial role in securing transactions within most cryptocurrency blockchains. It offers equivalent security with shorter key lengths compared to RSA; for instance, a 256-bit ECDSA key delivers similar security as a 2048-bit RSA signature [24]. Hence, the use of elliptic-curve tools in blockchains doesn't inherently protect against quantum attacks unless they align with PQC methods. The security of blockchains like Bitcoin relies on the honesty of the majority of network participants. If a dishonest node attempts to manipulate the system, the widely held consensus will ultimately reject its actions. Yet, a powerful node using significant computational resources could

execute an undetected double-spending attack by generating an invalid transaction, adding it to the blockchain, and creating a longer chain that includes the invalid transaction. As the longest chain is typically accepted by consensus, the double-spending transaction becomes valid, and the previous legitimate transaction is invalidated, resulting in a 51% attack. Quantum computers possess immense computational power compared to classical computers. For instance, with Grover's algorithm, a quantum computer could theoretically search through the vast Bitcoin hash rate, currently at 97 exahash per second, much faster than classical computers [6]. This speed-up implies that a malicious quantum machine could potentially achieve a 51% attack significantly faster than traditional computers, raising concerns about blockchain security.

Traditionally strength of public key cryptosystems' is measured in bits-of-security against classical computing attacks. This quantifies the effort needed for a traditional computer to execute a brute-force attack matching the key length. For example, a system is considered to have 1024-bit security if breaking it requires the same effort as brute-forcing a 1024-bit cryptographic key. Current 128-bit security cryptosystems are estimated to cost millions of dollars to break with classical computers and for the next few years, they are assumed to be secure. Quantum computers pose a significant threat. A 1000-qubit quantum system can break 160-bit elliptic curves [25], whereas a 1024-bit RSA system needs about 2000 qubits. As of now, the most powerful quantum computer (79 qubits) is not yet a practical threat. Even advanced organizations like the NSA haven't made substantial progress in building large quantum computers. However, it's projected that within the next 20 years, these computers will become capable of easily breaking current strong public key cryptosystems.

III. PQC ALGORITHMS

The fourth round at NIST PQC standardization involved assessment and discussions on the security levels of candidate algorithms including SPHINCS+ [7], FALCON [8], CRYSTALS-Kyber [9] and CRYSTALS-Dilithium [10] digital signature schemes and PKE/KEM algorithms as mentioned in Table 2, resulting in, standards set forth by NIST in August 2024 such as the Module-Lattice-Based Key-Encapsulation Mechanism Standard [21], Module-Lattice-Based Digital Signature Standard [22], and Stateless Hash-Based Digital Signature Standard [23]. Additionally, it emphasises the optimization measures to be employed, and the comparison of the

TABLE II
PARAMETER SUMMARY FOR NIST POST-QUANTUM ALGORITHMS

Parameter / Metric	FIPS 205 (SPHINCS ⁺)	FIPS 204 (Dilithium)	FIPS 203 (Kyber)
Artifact Size	Signatures ~40 KB	Signatures ~2 KB	Ciphertexts ~1 KB
Key Size	Large	Moderate	Very small
Security Level	~128-bit (Cat 2)	~192-bit (Cat 3)	~256-bit (Cat 5)
Computation Complexity	High (hash intensive)	Moderate (leveraging efficient NTT arithmetic)	Very low (optimized for resource-constrained devices)
Intended Application	UTXO-based systems	Smart contracts and PoS blockchains	IoT/DAG-based systems

results obtained with software implementations, effectiveness and resources needed to achieve this effectiveness for certain tasks and applications like IoT. Furthermore, projects such as IOTA [11], Winternitz One-Time Signature (W-OTS+) [12], Nexus [13], Cellframe [14], Hcash [15], Mochimo [16], QAN platform [17], Quantum Resistant Ledger [18], PQShield [19], and Quantum Blockchains project QSB (Quantum Secured Blockchain) [20] are also contributing to PQB. NewHope is a PQC algorithm and it is efficient and scalable. It protects against both classical and quantum attacks. BLISS is a PQC algorithm based on a lattice structure that offers strong protection against quantum attacks. eXtended Merkle Signature Scheme XMSS [26] is a PQC hash algorithm that offers very high security against quantum attacks. NTRU [27] is an example of a lattice-based PQC algorithm that is very safe from attacks from quantum computers. Crystal-Kyber has been optimized to be highly efficient therefore making it suitable for PQC applications where scalability is quite crucial. It also offers a high level of safeguard against attackers that use quantum techniques. The IOTA Foundation for instance is developing a signature scheme known as W-OTS+ that may be implemented on their blockchain. QAN is working on a hybrid solution to tackle the PQ paradigm. PQShield is a PQC firm based in the United Kingdom that is currently focusing on the creation of lattice-based security solutions for the blockchain. HashQuark in collaboration with the Chinese Academy of Sciences is working on a PQ secure blockchain platform. The project launched by Pulawy in 2018 called the Quantum Blockchain project is aimed at the further development of quantum-safe cryptography as a way to counteract potential threats.

IV. BLOCKCHAIN AND POST-QUANTUM CRYPTOGRAPHY

A. Bitcoin

Satoshi Nakamoto introduced Bitcoin in 2008 as the pioneering peer-to-peer digital cash system, revolutionizing the financial landscape. It ingeniously addressed the double spending problem through a decentralized mechanism reliant on a distributed timestamp server. Bitcoin's core infrastructure is built on blockchain technology in which blocks are securely connected using cryptographic hashing. Each block serves as an immutable ledger, housing a permanent record of transactions. The inclusion of hash ensures tamper-proofing through visibility and integrity. Bitcoin maintains a block time of 10

minutes, with a new block added to the blockchain approximately every 10 minutes. This temporal consistency stays sustained by the complexity that is levied by the Proof-of-Work (PoW) mechanism. Miners employ SHA-256 hashing, along with a random number called a nonce, to certify that the resulting hash value falls below a specific target. This target is adjusted roughly every two weeks to sustain the 10-minute block time [6]. To demonstrate rights to a Bitcoin in a transaction, the coin's holder must sign the transaction using their corresponding private key. This signature process invalidates the spent coin while creating a new one with the same value, designated for the recipient's public key, establishing them as the new owner. Bitcoin's transaction security depends on ECDSA, specifically employing the Secp256-k1 elliptic curve.

Bitcoin's security is currently under scrutiny due to its vulnerability to potential quantum attacks. Shor's algorithm, when implemented on a quantum machine, is capable of decrypting ECDSA in polynomial time $O(n^3)$ [24], raising concerns about the cryptocurrency's long-term resilience due to the continuously advancing quantum era.

B. Ethereum

Ethereum utilizes a modification of the ECDSA method built on the "secp256-k1 elliptic curve". In Ethereum transactions, the main public key linked with an account is not overtly disclosed due to the absence of a "from" field [6]. Ethereum embarked on a transition from its original PoW consensus mechanism, which was implemented when it was launched in 2015, to a Proof-of-Stake (PoS) system in Ethereum 2.0. This new system's security relies on the principle that miners with a higher stake in the network gain more voting power. Users are incentivized to act honestly by staking more coins since they risk losing more if they engage in malicious behaviour. Despite the transition to Ethereum 2.0 and the PoS mechanism, the signature structure remains susceptible to Shor's algorithm-based quantum attacks. This vulnerability persists because the signature scheme is rooted in the DLP, which is recognized to be susceptible to quantum computing advancements. Ethereum 3.0 takes a significant leap forward in quantum-resilient features by introducing support for multi-signatures that remain impervious to quantum threats. This development includes plans for the integration of PQC elements such as zkSTARKs. Ethereum 3.0 also embraces zk-SNARKs "Zero-Knowledge Succinct Non-Interactive Argument of Knowledge", relying on a secure system that necessitates the creation of a larger

public key. Importantly, this global public key lacks a corresponding private key, a precautionary measure designed to thwart unauthorized coin generation, akin to what's observed in Zcash [6]. The creation of this expanded public key entails a collaborative effort among multiple users, each contributing small segments generated from their respective private keys. Notably, if one individual were to lose their private key, deciphering the private key corresponding to the larger public key becomes computationally insurmountable on conventional computers. This is because uncovering this private key would require solving the DLP or gaining access to all the private keys employed in generating the larger public key. With these advancements, Ethereum 3.0 stands as a pivotal platform in the PQC landscape, well-equipped to accommodate smart contracts in the forthcoming quantum world. Furthermore, such a transition extends quantum resistance towards numerous cryptocurrencies operating on the Ethereum ERC-20 platform.

C. Monero

Monero employs a concealed public ledger to ensure transaction traceability remains elusive. The use of Ring Confidential Transaction (RingCT) obscures both the transaction amount and the sender's public key. Transactions are initiated and transmitted through any network peer, rendering their sources indistinguishable from external observers. Monero's ring digital signature scheme permits multiple peers to sign a transaction. Once a sender initiates a transaction, they sign it with their key along with ten other keys, resulting in a total of 11 input keys potentially associated with the signed transaction. This approach introduces obfuscation and renders transactions untraceable [6]. Monero employs an altered EdDSA, relying on the DLP. These signatures are vulnerable to quantum attacks by Shor's algorithm. Despite the vulnerability of EdDSA, Monero's use of RingCT necessitates a quantum attacker to resolve several Pedersen commitments on the way to identify the accurate public key that is implemented in a transaction, hence adding complexity to the attack, and making it considerably more challenging. Monero's consensus protocol update, incorporating RandomX, enhances resistance against quantum attacks. It fortifies the network against potentially Grover's algorithm-based 51% attacks [24], providing an additional layer of security.

D. Zcash

Zcash places a strong emphasis on ensuring the absolute privacy of its transactions through the use of the powerful tool of Zero-knowledge proofs, which allows the verification of transaction validity without disclosing any details about the sender, receiver, or transaction amount [6]. Zcash also provides selective disclosure capabilities, allowing users to choose to reveal certain transaction details, if necessary, typically for compliance or audit purposes. This feature enables a level of flexibility while maintaining privacy. Recently a new version of Zcash became accessible on the Ethereum blockchain over a procedure referred to as "wrapping." This integration made Zcash well-suited with key wallets and Dapps on the Ethereum

platform, expanding its usability. Zcash employs ECDSA in conjunction with zk-SNARKs for its cryptologic operations [24]. Yet, it is imperative to acknowledge that this cryptographic setup is vulnerable to quantum attacks, specifically the application of Shor's algorithm. This susceptibility raises concerns regarding the security of Zcash's signature scheme. Beyond the vulnerabilities in its signature scheme, Zcash faces a significant threat in the form of quantum attacks on its consensus process [6]. This scenario could potentially lead to a quantum-powered 51% attack on the network, posing a substantial risk to its integrity. Table 3 gives a summary of the blockchain characteristics and quantum vulnerabilities.

V. ANALYSIS OF BLOCKCHAIN TRANSITION TO PQC

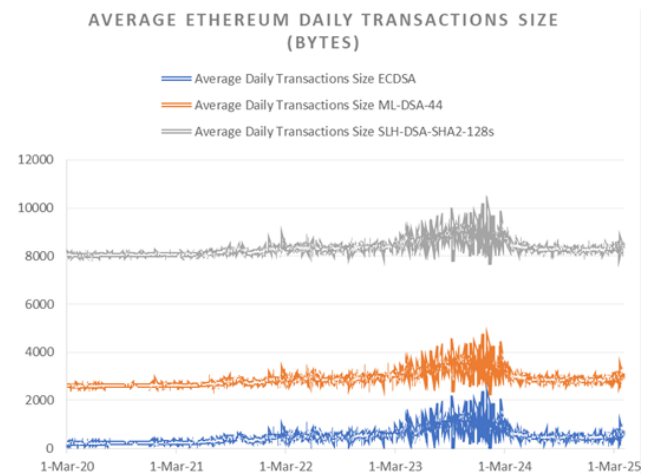


Fig. 1. Average Ethereum Daily Transactions Size (Bytes)

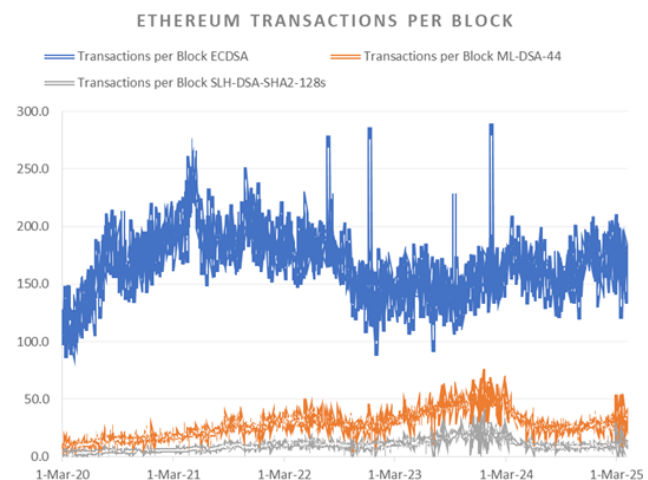


Fig. 2. Ethereum Transactions Per Block

To evaluate the significance of post quantum cryptographic algorithm, we will be focusing on the last 5 years of Ethereum

TABLE III
SUMMARY OF BLOCKCHAIN CHARACTERISTICS AND QUANTUM VULNERABILITIES

Blockchain	Consensus Mechanism	Signature Scheme	Quantum Vulnerabilities	Remarks / PQC Transition
Bitcoin	Proof-of-Work (10-minute block time)	ECDSA (Secp256k1, SHA-256)	Vulnerable to Shor's algorithm (can break ECDSA)	Established UTXO model; full transition is challenging due to legacy state and protocol rigidity
Ethereum	Transitioning from PoW to PoS (Ethereum 2.0/3.0)	Modified ECDSA (Secp256k1); future PQC multi-signatures (e.g., incorporating zk-STARKs)	Vulnerable to quantum attacks on discrete logarithm problems	PQC upgrades planned (e.g., multi-signatures and collaborative public key expansion) via soft-fork updates
Monero	Proof-of-Work with privacy enhancements (RandomX, RingCT)	Ring signatures (altered EdDSA)	Vulnerable to quantum attacks (Shor's algorithm on DLP), though obfuscation via multiple keys increases complexity	Privacy-focused design; additional measures help mitigate quantum threats but signers remain at risk
Zcash	Proof-of-Work (with recent Ethereum wrapping)	ECDSA with zk-SNARKs for privacy	Vulnerable to Shor's algorithm affecting ECDSA; additional consensus risks from quantum 51% attacks	Emphasizes strong privacy; transition to a quantum-secure model will require substantial protocol updates

transactions data from etherscan.io [28] and two of the NIST standardized post quantum signature algorithms FIPS 204 Module-Lattice-Based Digital Signature Standard (ML-DSA-44) and FIPS 205 Stateless Hash-Based Digital Signature Standard (SLH-DSA-SHA2-128s). The purpose of focusing on the last 5 years of Ethereum data is to better understand a stable infrastructure with considerable user base. It is important to note that FIPS 203 is a key encapsulation algorithm while FIPS 204 and 205 are digital signature algorithms and even though these algorithms are not specifically designed for blockchain applications, it is important to analyse the threshold of improvements and modifications required to possibly migrate blockchain to PQC. Figure 1 shows the Signature Impact on Average Ethereum Daily Transactions Size (Bytes) and Figure 2 represents the Signature Impact on Transactions Per Block. The figures help us visualise the impact the latest NIST PQC standards ML-DSA-44 and SLH-DSA-SHA2-128s will have as compared to the traditional ECDSA cryptographic algorithm. There are multiple performance implications of this upgrade, including a significant increase in the transaction size. The throughput drops from 15 TPS for ECDSA to approximately 2.7 TPS (5.5x slower) for ML-DSA-44 and less than 1 TPS for SLH-DSA-SHA2-128s. This indicates severe consequences for network usability, economics, and decentralisation. Instead of handling around 180 transactions per block, it drops to just 33 for ML-DSA-44 or even 11 for SLH-DSA-SHA2-128s. Ethereum's transaction capacity could decrease significantly, potentially reducing its throughput from 15 to less than 3 TPS a level comparable to Bitcoin's base layer, impacting its ability to support decentralised applications, financial protocols, and everyday user activity. This could lead to increased competition for block space, increasing gas fees, and potentially increasing the cost of everyday transactions, alienating retail users and stifling innovation.

VI. CONCLUSION

The rise of quantum computing will be a major risk to blockchain network security. However, through the development of PQB technology, we can mitigate these risks and ensure the continued success of blockchain in the future. Through the incorporation of PQC algorithms, we fortify the security, integrity, and privacy of blockchain data, establishing a persistent infrastructure for forthcoming applications and services. While the path to achieving PQB remains a work in progress, collaborative endeavours uniting researchers, developers, and industry pioneers hold the promise of crafting a durable blockchain ecosystem for the quantum age. Selecting a PQB cryptosystem is a complex decision influenced by factors such as blockchain node hardware, available resources (memory and speed), required performance, and desired security levels. However, it's essential to recognize that results are hardware-specific, and performance may vary considerably on different platforms. Beyond the discussed cryptosystems, it's vital to explore additional security features that have seen limited use in non-academic blockchain projects. These features include secure multi-party computation, ring or aggregate signatures, homomorphic or identity-based encryptions, zk-SNARKs, and secret sharing. These features can enhance security and privacy in blockchain applications. New blockchains should refrain from using the vulnerable ECDSA and instead opt for PQC solutions in their design. PQB solutions must not rely on a particular PQC signature scheme, rather they should be designed to be scheme-independent to ensure long-term security and adaptability to algorithmic advances. Novel PQBs ought to consider alternative consensus algorithms and should avoid PoW to decrease the possible risk of 51% attacks by quantum computers. While the article primarily focuses on blockchain, the recommendations and conclusions can be applied to other Distributed Ledger Technologies (DLTs) with similar operation methods. Directed Acyclic

Graphs (DAGs) based DLTs or Hashgraphs can also benefit from these insights. There is a significant need to upgrade the Blockchain to post-quantum cryptography to tackle the challenges of the future post-quantum threat. Researchers have to rigorously recalibrate and modify NIST standardized post quantum signature algorithms FIPS 204 Module-Lattice-Based Digital Signature Standard (ML-DSA-44) and FIPS 205 Stateless Hash-Based Digital Signature Standard (SLH-DSA-SHA2-128s), so they can be implemented on resource-constrained and time-sensitive Blockchain infrastructures to maintain the throughput and performance standards expected from blockchain transactions.

ACKNOWLEDGEMENT

This work was supported in part by project SERICS (PE00000014) under the NRRP MUR program funded by the EU - NGEU. Views and opinions expressed are however those of the authors only and do not necessarily reflect those of the European Union or the Italian MUR. Neither the European Union nor the Italian MUR can be held responsible for them.

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